

Session 1: Introduction to Financial Analytics & Regression Foundations

From Business Questions to Boardroom Insights — Building Analytical Intuition

DATA & CONTEXT

SVB collapse, financial news & why analytics matters now

REGRESSION CORE

OLS mechanics, coefficient intuition & R^2 interpretation

LIVE CODING

Stata & Python demos on 8-bank benchmark dataset

What We Cover Today — Session 1 Architecture

Module 1

The Business Case for Analytics

Why analytics? SVB case, four levels, the workflow.

Module 2

Data Architecture

Cross-sectional, time-series, panel data & sourcing.

Module 3

Regression Foundations

Population model, OLS, intercept, slope, R^2 .

Module 4

OLS Assumptions & Diagnostics

8 assumptions, diagnostic tools, what violates what.

Module 5

Live Coding Demos

Stata & Python: import → visualise → estimate → diagnose.

Module 6

Group Challenge

Build and defend your own model in 20 minutes.

Silicon Valley Bank: \$209 Billion — Gone in 48 Hours



What Happened? The Speed of a Modern Bank Run

- SVB held \$91B in long-duration US Treasury bonds.
- When interest rates rose, bond prices FELL — creating unrealised losses.
- Depositors (tech founders, VCs) got nervous and withdrew \$42B IN ONE DAY.
- SVB collapsed within 48 hours — fastest bank failure in US history.

Why Didn't Analysts See This Coming?

- Duration mismatch was **VISIBLE** in plain regulatory filings.
- A basic regression of bank equity sensitivity to rate rises would have flagged massive vulnerability.



CLASS DISCUSSION (2 Min)

- Q1: What data would you need to predict SVB's vulnerability?
- Q2: What was the **DEPENDENT** variable an analyst should model?
- Q3: What were the key explanatory variables (X) in this case?

Why Financial Analytics? Four Managerial Urgencies



Complexity of Modern Finance

Markets generate terabytes of data daily. Human intuition alone cannot process signals vs noise.



Speed of Capital Decisions

Algorithmic trading, real-time credit scoring, and HFT demand model-driven decisions in milliseconds.



Regulatory Accountability

Basel III, RBI SREP, SEBI disclosures — all require quantitative risk models and stress-test evidence.



Competitive Advantage

JPMorgan, BlackRock, Goldman Sachs run thousands of regressions daily. Your competition is already quantitative.

The Four Levels of Financial Analytics

1. DESCRIPTIVE

'What happened?'

Summarise historical financial data. Dashboard KPIs: NPA trends, revenue growth, credit spreads.

2. DIAGNOSTIC

'Why did it happen?'

Drill down into root causes. Regression: What drove NPA spike? Which borrower segments defaulted?

3. PREDICTIVE

'What will happen?'

Forecast future outcomes. Machine learning: Credit scoring, default probability, loan demand forecasting.

4. PRESCRIPTIVE

'What should we do?'

Optimise decisions. Portfolio optimisation, dynamic pricing, regulatory capital allocation.

The Financial Analytics Workflow: 6-Step Process

1 FRAME

Define the managerial question precisely.

2 DATA

Identify, collect & clean relevant data.

3 EXPLORE

EDA: distributions, correlations, outliers.

4 MODEL

Specify & estimate econometric model.

5 DIAGNOSE

Test assumptions; fix violations.

6 COMMUNICATE

Translate results into business decisions.

Data Architecture: Three Types of Financial Data

Cross-Sectional

Snapshot of multiple entities at ONE point in time.

Example: NIM, NPA ratio, CAR for 50 NSE-listed banks on 31-Mar-2024.

Challenge: Cannot track change over time. Unobserved firm heterogeneity.

Time-Series

One entity observed REPEATEDLY over time.

Example: Monthly RBI repo rate (2000–2024), NIFTY 50 daily returns.

Challenge: Autocorrelation, stationarity, structural breaks during crises.

Panel Data

Multiple entities observed repeatedly over time (BEST of both).

Example: Quarterly ROA, NPA, capital ratios for 40 banks over 10 years.

Gold Standard: Controls for unobserved bank-specific fixed effects!

From Population to Sample — The Regression Line

Population (Unknown Reality)

$$Y = \beta_0 + \beta_1 X + u$$

- β_0 = True intercept
- β_1 = True slope
- u = All other influences

These CANNOT be observed directly.

We can NEVER know the true β_1 with certainty!

Sample (What We Estimate)

$$\hat{Y} = b_0 + b_1 X$$

- b_0 = Estimated intercept
- b_1 = Estimated slope
- $e = Y - \hat{Y}$ (residual)

OLS minimises $\sum e^2$ to find b_0, b_1 .

Q: If we took a different sample of 8 banks, would we get the same b_1 ? Why or why not?

Actual vs. Predicted — Reading the Regression Line

8-Bank Scatter: ROA vs Bank Size

Bank	ROA (%)	Size (₹ bn)	Predicted $\hat{Y}$	Residual
A	0.7	120	0.814	-0.114
B	1.0	180	0.916	+0.084
C	1.2	260	1.052	+0.148
D	0.8	300	1.120	-0.320 ←
E	1.5	420	1.324	+0.176
F	1.7	520	1.494	+0.206
G	1.3	610	1.647	-0.347 ←
H	1.9	750	1.885	+0.015

Estimated Line: $\hat{Y} = 0.61 + 0.0017 \times \text{Size}$ | $R^2 = 0.74$

? CLASS QUESTIONS

- Which bank has the largest residual?
- Bank D: well-explained by size alone?
- Which bank is most OVER-predicted?
- What omitted factor could explain Bank D's low ROA despite large size?

Understanding the Intercept — When Does 'Zero' Make Sense?

🤔 If bank SIZE is zero — does a bank with zero assets make any business sense?

Definition Predicted Y when $X = 0$.

Finance Example Predicted loan growth when GDP growth = 0%.

Meaningful Case Baseline credit score when borrower income = 0 (floor score).

Meaningless Case Predicted ROA when size = ₹0 billion — no such bank exists.

Rule of Thumb Interpret the intercept ONLY when $X = 0$ is a plausible, real-world reference point.

📌 As a finance manager: Focus your interpretation on the SLOPE. The intercept is usually a mathematical anchor, not a business insight.

Understanding the Slope — The Heart of the Model

$$b_1 = \Delta \hat{Y} \div \Delta X \quad (\text{estimated change in } Y \text{ per ONE-unit change in } X)$$

Sign → Direction

Positive: X and Y move together. Negative: they move oppositely. Ask: If $b_1 > 0$, does higher leverage → higher or lower ROA?

Magnitude → Economic Size

A b_1 of 0.001 is very different from 0.5 — both are 'one unit' changes. Always ask: is this economically meaningful, not just significant?

Units Matter Enormously

If X is in ₹ crore vs ₹ billion, b_1 changes by a factor of 100. State units explicitly in EVERY table, every time.

Context Example

$b_1 = 0.65$ on GDP growth (%): A 1 percentage point rise in GDP growth → 0.65 ppt higher loan growth, on average.

Prediction From a Regression Equation

$$\hat{Y} = 1.50 + 0.65 \times X \quad (X = \text{GDP growth in \%})$$

$$X = 2\% \rightarrow \hat{Y} = 1.50 + 0.65 \times 2 = 2.80\%$$

$$X = 5\% \rightarrow \hat{Y} = 1.50 + 0.65 \times 5 = 4.75\%$$

$$\text{Change (2\%} \rightarrow \text{5\%)} \rightarrow \Delta\hat{Y} = 1.95 \text{ ppt}$$

Rule:

Each additional 1 ppt of GDP growth adds 0.65 ppt of Loan Growth — regardless of starting level.

Managerial Use:

RBI projects GDP at 6.5%. Stress-test: If GDP falls to 4.5%, model predicts loan growth falls by 1.30 ppt.



IN-CLASS CALCULATION (90 secs)

- Using $\hat{Y} = 1.50 + 0.65 \times X$, calculate predicted loan growth when GDP = 3.5%.
- By how much does predicted loan growth change if GDP improves from 2% to 4%?
- Answer: $\hat{Y} = 3.775\%$ | Change = $0.65 \times 2 = 1.30$ ppt increase.

What Does OLS Actually Do? — Intuition Before Formula

"Among all possible straight lines you could draw through your data, OLS picks the one that minimises the total squared vertical gaps between actual and predicted values."

Step 1 Calculate the vertical gap for each bank: Actual ROA – Predicted ROA. This is the residual e .

Step 2 Square each residual. Why? Positive and negative errors cannot cancel each other out.

Step 3 Sum up all the squared residuals: $\sum e^2 = \sum (Y - \hat{Y})^2$.

Step 4 OLS finds the values of b_0 and b_1 that make this total sum as small as possible.

Step 5 That is the 'best fit' line — it minimises total squared errors across all data points.

Residuals — The Model's Confession of What It Missed



Residual Table: GDP Growth vs Loan Growth

GDP Growth (%)	Actual Y (%)	Predicted $\hat{Y}$ (%)	Residual e
1.0	2.10	2.15	-0.05
2.0	2.70	2.80	-0.10
3.0	3.20	3.45	-0.25
4.0	4.40	4.10	+0.30
5.0	5.20	4.75	+0.45 ←
Largest			



THINK–PAIR–SHARE (2 mins)

- Which observation has the largest absolute residual?
- At GDP growth = 5%, actual loan growth was 5.20% but predicted only 4.75%.
- What economic forces explain this under-prediction?
- What variable is missing from the model?

R^2 — Useful, Loved, and Dangerously Misused

R^2 = Proportion of variation in Y explained by the model = ESS / TSS



What R^2 TELLS you

How much of the variation in Y is captured by your included X variables.



Example — $R^2 = 0.60$

60% of variation in bank ROA is explained by size, leverage, and liquidity.



What R^2 does NOT tell you

Whether the relationship is causal. Or whether the model is well-specified.



High $R^2 \neq$ good model

You can get $R^2 = 0.95$ with a completely wrong set of variables.



Low $R^2 \neq$ useless model

In cross-sectional finance, R^2 of 0.15 can still carry meaningful and robust coefficients.

Simple vs. Multiple Regression — When to Use Each

Simple Regression (Bivariate)

$$Y = \beta_0 + \beta_1 X + u \quad (\text{one } X \text{ variable})$$

Finance Example:

$$\text{ROA} = \beta_0 + \beta_1 \times \text{Size} + u$$

- Intuition-building: see raw bivariate relationship.
- Easy to visualise on a scatter diagram.
- Baseline reference before adding controls.

⚠ Limitation: Omitted variable bias. If other drivers of ROA are correlated with Size, the slope will be biased.

Multiple Regression (Multivariate)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + u$$

Finance Example:

$$\text{ROA} = \beta_0 + \beta_1 \text{Size} + \beta_2 \text{Leverage} + \beta_3 \text{Liquidity} + u$$

- Controls for confounders simultaneously.
- Ceteris paribus interpretation ('all else equal').
- Reduces omitted variable bias.

✅ Goal: Isolate the effect of each X on Y holding others constant.

Simple Regression Result — ROA on Bank Size

 **Prediction Moment:** What sign do you expect on the size coefficient?

Estimated Equation:

$$\hat{Y} = 0.61 + 0.0017 \times \text{Size} \quad (\text{Size in ₹ billion})$$

$$R^2 = 0.74$$

How to Interpret:

- Intercept (0.61): Baseline ROA when Size = 0 (not meaningful here).
- Slope (+0.0017): Each ₹1 billion increase in bank size is associated with a +0.0017 ppt higher ROA.

Economic translation:

₹100 billion more in assets → ROA rises by 0.17 percentage points.

 **DISCUSS (2 mins)**

- Is the 0.0017 slope economically meaningful?
- What does $R^2=0.74$ tell us here?
- What omitted variable would you add next?
- Does SIZE cause higher ROA — or does high ROA allow banks to grow?

Why Simple Regression May Mislead

"A single-variable model is like a doctor diagnosing with one symptom. Sometimes it works. Often it misses the point."

Real Problem

Bank profitability has MANY drivers — size is just one of them.

The simple model forces the SIZE coefficient to absorb ALL omitted effects.

Confounding

Larger banks may also tend to have LOWER leverage and MORE liquidity.

Without controlling for these, the size coefficient picks up their effects too.

OVB Direction

If high-leverage banks are systematically SMALLER, β_1 is biased UPWARD.

We overestimate the true benefit of scale.

Solution

Add leverage, liquidity, and macro variables as controls.

Multiple regression fixes OVB by holding confounders constant.

Omitted Variable Bias — Think Like an Analyst

⚠ OVB Formula

Model: $ROA = \beta_0 + \beta_1 \text{Size} + u$ (Leverage is OMITTED)

OVB Formula: $E(b_1) = \beta_1 + \beta_2 \times [\text{Cov}(\text{Size}, \text{Leverage}) / \text{Var}(\text{Size})]$

Two Conditions: 1) Omitted variable must cause Y ($\beta_2 \neq 0$) 2) Omitted variable must be correlated with X1 [$\text{Cov}(X1, X2) \neq 0$]

Short Model (Biased)

$$ROA = 0.50 + 0.20 \times \text{Size} + e$$

Slope looks strongly positive.

Conclusion: Bigger banks are always more profitable?

Full Model (Unbiased)

$$ROA = 0.40 + 0.08 \times \text{Size} - 0.12 \times \text{Leverage} + e$$

Size slope drops from 0.20 to 0.08!

Conclusion: Size benefits are partly explained by leverage differences.

Adding More Instruments to the Orchestra

Multiple regression lets us hold other factors constant — and isolate the effect of any single driver.

Our Bank Model

$$\text{ROA} = \beta_0 + \beta_1 \text{Size} + \beta_2 \text{Leverage} + \beta_3 \text{Liquidity} + u$$

- β_1 = Effect of Size on ROA, holding Leverage and Liquidity CONSTANT.
- β_2 = Effect of Leverage on ROA, holding Size and Liquidity CONSTANT.
- β_3 = Effect of Liquidity on ROA, holding Size and Leverage CONSTANT.

Estimated Multiple Regression Result

$$\text{ROA}_{\text{hat}} = 0.80 + 0.003 \times \text{Size} - 0.15 \times \text{Leverage} + 0.02 \times \text{Liquidity}$$

Interpretation: Each ₹1 bn more in assets → +0.003% ROA, for banks with identical leverage and liquidity.

The Ceteris Paribus Principle — 'All Else Equal'

Ceteris Paribus = 'all other things being equal / held constant' — the two Latin words that carry the entire logic of multiple regression.

When we say $\beta_1 = 0.003$ (size coefficient), we mean:

"Comparing two banks that have the SAME leverage and SAME liquidity — the one that is ₹1 billion larger in assets is predicted to have 0.003% higher ROA."

Finance Examples:

- $\beta_{\text{Leverage}} = -0.15 \rightarrow$ Two banks with same size and liquidity: the one with 1-unit higher D/E ratio has -0.15% lower ROA.
- $\beta_{\text{ESG}} = +2.5 \rightarrow$ Two firms of same size and sector: the one with 1-point higher ESG score has +2.5% higher Tobin's Q.

⚠ Never interpret a multiple regression coefficient as if the other variables did not exist in the model!

Simple vs. Multiple Regression — Head to Head

Dimension	Simple Regression	Multiple Regression
X Variables	One predictor	Two or more predictors
Interpretation	Raw bivariate association	Partial association (ceteris paribus)
Omitted Variable Bias	HIGH — other drivers absorbed into slope	LOWER — controls reduce confounding
Complexity	Simple, easy to visualise	Richer, harder to visualise
Finance Use Case	Baseline exploration and quick EDA	All serious decision-support models

OLS Assumptions — Why Should an MBA Student Care?

 **Real-world case: A major Indian bank used a credit scoring model with a violated zero-mean error assumption. The model performed brilliantly in training — and caused a ₹3,200 Crore NPA surge after deployment.**

A1 **Linear in Parameters** Model must be linear in coefficients (not necessarily in X variables).

A2 **Random Sampling** Data must represent the target population — no survivorship or selection bias.

A3 **No Perfect Multicollinearity** No X variable is an exact linear combination of others.

A4 **Zero Conditional Mean** $E(u|X) = 0$ — the KEY assumption for unbiasedness.

A5 **Homoskedasticity** $\text{Var}(u|X) = \sigma^2$ constant across all observations.

A6 **No Serial Correlation** Errors must not be correlated across time periods.

Linear in Parameters — More Flexible Than You Think

The model must be linear in the UNKNOWN COEFFICIENTS — not necessarily in X. This is far more flexible than students think!

✓ $Y = \beta_0 + \beta_1 X + u$

Valid

Classic linear form. Standard model.

✓ $Y = \beta_0 + \beta_1 \log(X) + u$

Valid

Log-linear: one X is logged. Still linear in β .

✓ $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + u$

Valid

Quadratic: adds squared X. Still linear in β .

✗ $Y = \beta_0 + \beta_1^X + u$

Invalid

β appears as an exponent — non-linear in parameters.

✗ $Y = \beta_0 \times \beta_1^{X_1} \times \beta_2^{X_2}$

Invalid

Multiplicative parameters — requires transformation.

Sampling & Variation in X

Assumption 2: Appropriate Sampling

- The sample must represent the target population.
- Selection bias: who is NOT in your database?
- Survivorship bias: only analysing CURRENTLY listed NSE firms ignores the firms that went bankrupt and delisted.

Example:

Mutual fund return analysis using only 2024-listed funds overstates performance by excluding failed funds.

Assumption 3: Variation in X

- X must not be constant — we cannot estimate slope from a single X value.
- No perfect collinearity between X variables.

Example:

If all banks in your sample have IDENTICAL Tier-1 capital ratios (e.g., SEBI regulation forces them to 12%), there is ZERO variation — and OLS cannot estimate β_1 for CAR.

No Perfect Multicollinearity — Separating the Effects

"Imagine asking two witnesses who were together the entire time: 'Who did it?' They cannot tell you. That is the multicollinearity problem."

Perfect Multicollinearity

One X is an EXACT linear combination of others.

OLS fails completely — cannot compute estimates.

Example:

Including both 'Debt' and 'Debt/Assets Ratio' in the same model — one is a mathematical transformation of the other.

Imperfect Multicollinearity

X variables are HIGHLY but not perfectly correlated.

OLS still works — but standard errors inflate, making t-statistics unreliable.

Diagnosis: Correlation matrix + VIF (Variance Inflation Factor).

Fix: Drop one variable, combine into an index, or use PCA.

Zero Conditional Mean — The King of Assumptions

$E(u | X) = 0$ ← The error term must not systematically move with X. This is the most important assumption in all of econometrics.

In Plain English:

There should be no pattern left in the error term that can be predicted from X.

If $E(u|X) \neq 0$, then b_1 is BIASED — OLS coefficient does not recover the true β_1 !

Three Common Violations:

- **Omitted Variable Bias:** Unobserved factors (managerial quality, bank culture) that correlate with X and cause Y.
- **Reverse Causality:** Does bank profitability (Y) cause banks to GROW (X)? Or does size cause profits?
- **Measurement Error:** If X (e.g., 'announced' credit rating) is measured with error correlated with u.

Fix: Instrumental Variables (IV), Difference-in-Differences (DiD), or lagged variables.

Homoskedasticity — Are Your Errors Well-Behaved?

"Imagine forecasting weather: for summer, off by $\pm 2^\circ\text{C}$. For monsoon, off by $\pm 8^\circ\text{C}$. Your forecast errors are not constant — that is heteroskedasticity."

✓ Homoskedasticity

- Constant error variance: $\text{Var}(u|X) = \sigma^2$
- Standard errors are reliable.
- t-statistics and p-values are valid.
- OLS is BLUE (Best Linear Unbiased Estimator).

Residual vs Fitted Plot: Errors scatter evenly around zero — a random horizontal band.

✗ Heteroskedasticity

- $\text{Var}(u|X) = \sigma_i^2$ (changes with X!)
- Finance Example: Small NBFC errors are tiny; Reliance earnings errors are massive.
- Standard errors are **WRONG** — usually understated.
- False t-statistics → Spurious significance!

Fix: Huber-White Robust Standard Errors (HC1).

Serial Correlation & Normality

Assumption 7: No Serial Correlation

- Errors must not be correlated over time.
- Critical in time-series and panel data.

Example:

If NIFTY falls sharply today, panic often spills into tomorrow — creating correlated errors.

Effect: Conventional SEs are too small → overconfident inference.

Diagnosis: ACF plots, Durbin-Watson, Breusch-Godfrey test.

Fix: Newey-West SEs, ARIMA, time fixed effects.

Assumption 8: Normality of Errors

- Errors $u \sim \text{Normal}(0, \sigma^2)$ distribution.
- Useful for exact small-sample inference.
- NOT required for OLS unbiasedness.

Large Sample?

Central Limit Theorem (CLT) saves us — even without normality, sampling distributions of b_1 converge to normal.

Diagnosis: Histogram of residuals, Q-Q plot.

Fix: Log-transform Y; remove extreme outliers.

The Diagnostic Map — Assumption → Question → Tool

Issue / Assumption	Ask First...	Diagnostic Tools	If Violated...
Functional Form	Is the relationship misspecified?	Scatter plots, RESET test	Respecify X or use log/quadratic
Multicollinearity	Can X effects be separated?	Correlation matrix, VIF > 5	Drop/merge variables or PCA
Heteroskedasticity	Does error variance change with X?	Residual plot, Breusch-Pagan	Use HC-robust standard errors
Serial Correlation	Do errors persist over time?	ACF plot, DW, BG test	Newey-West SEs, ARIMA
Exogeneity (ZCM)	Is X correlated with u?	Economic theory + IV diagnostics	IV estimation, DiD, RDD

Import, Inspect, Describe — Always Start Here

Before running a single regression, know what the data looks like. This is non-negotiable.

Stata: Import & Describe [STATA]

```
import excel "bank_data.xlsx", firstrow clear
```

* Step 1: Understand the structure

```
describe
```

* Step 2: Summary statistics

```
summarize roa size leverage liquidity
```

* Step 3: Check for missing values

```
misstable summarize
```

* Step 4: Codebook for details

```
codebook roa size
```



PREDICT BEFORE RUNNING

- What will describe output show?
- How many observations? Any missing values?
- Will all variables have plausible ranges?
- After output: Is mean ROA reasonable for Indian banks?
- Are min/max values plausible?

Visualise and Run Regression — Live Estimation

Stata: Visualise & Estimate [STATA]

* Scatter plot with fitted line

```
scatter roa size || lfit roa size
```

* Simple regression

```
reg roa size
```

* Multiple regression

```
reg roa size leverage liquidity
```

* Robust standard errors

```
reg roa size leverage liquidity, vce(robust)
```



PREDICTION MOMENT

- Before running — predict the sign of each coefficient:
 - Size (₹ bn): + or –? (Economies of scale vs too-big-to-manage)
 - Leverage: + or –? (Higher debt → more risk → lower ROA?)
 - Liquidity (%): + or –? (More cash → lower yields or better solvency?)
- After output: Did your predictions match? Which surprised you?

Basic Diagnostics — Checking the Model's Health

The output table is not the end — it is the beginning. Now we check whether to trust it.

Stata: Diagnostics [STATA]

```
* After OLS estimation:
predict yhat, xb          // fitted values
predict resid, residuals  // residuals
* Residual vs Fitted plot
rvfplot, yline(0)
* VIF check (multicollinearity)
estat vif
* Breusch-Pagan het test
estat hettest
* Distribution of residuals
histogram resid, normal
```

`rvfplot` No pattern = good. Funnel shape = heteroskedasticity!

`estat vif` All VIF < 10. Any > 10 = multicollinearity warning.

`estat hettest` $p > 0.05$: homoskedasticity not rejected. $p < 0.05$: use robust SEs.

`histogram resid` Bell-shaped = normality assumption plausible.

Load and Explore Data — Python Workflow

Python is increasingly the preferred tool in industry — especially for large datasets and ML integration.

Python: Load & Explore [PYTHON]

```
import pandas as pd
import matplotlib.pyplot as plt
# Load the dataset
bank = pd.read_excel('bank_data.xlsx')
# Step 1: First look
print(bank.head())
# Step 2: Data types and structure
print(bank.info())
# Step 3: Summary statistics
print(bank.describe())
# Step 4: Correlation matrix
print(bank[['roa', 'size', 'leverage', 'liquidity']].corr())
```

`head()` First 5 rows — sanity check

`info()` Data types, missing counts

`describe()` Summary stats for all columns

`.corr()` Pairwise correlation matrix

`isnull().sum()` Count missing values per column

Plot and Estimate OLS — Statsmodels in Action

Python: OLS Estimation [PYTHON]

```
import statsmodels.api as sm

# Scatter plot with regression line
bank.plot.scatter(x='size', y='roa')
plt.show()

# Simple regression: ROA on Size only
X1 = sm.add_constant(bank['size'])
m1 = sm.OLS(bank['roa'], X1).fit()
print(m1.summary())

# Multiple regression + robust SEs
X2 = sm.add_constant(bank[['size', 'leverage', 'liquidity']])
m2 = sm.OLS(bank['roa'], X2).fit(cov_type='HC1')
print(m2.summary())
```

coef	Estimated β — focus on sign AND magnitude
------	---

std err	Uncertainty of estimate (smaller = better)
---------	--

t	t-statistic = coef / std err ($ t > 2$ → significant)
---	---

P> t	p-value: <0.05 = significant at 5% level
------	--

R-squared	Proportion of Y variation explained by model
-----------	--

 Q: Will R^2 be higher in the multiple regression model?	35 / 37
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Build and Defend a Model — Real Analyst Work

This is what analysts actually do. Work in groups of 4. You have 15 minutes. Then present to the class.

1 Define the DV

What is your dependent variable? Why is it the best measure of bank profitability?

2 Choose 3 X Variables

Select three explanatory variables from the dataset. Why these three?

3 Predict Signs

Before running: what sign do you expect on each coefficient? Write it down.

4 Omitted Variables

Name one variable you cannot observe but suspect matters for ROA.

5 Reverse Causality

For one of your X variables: does Y potentially cause X instead?

6 Model Choice

Would you use simple or multiple regression? Defend in one sentence.

Takeaways — What You Now Know That You Didn't Before

- 1 Financial analytics begins with a clearly framed question and a correctly structured dataset.
- 2 Cross-sectional, time-series, and panel data each require different analytical thinking.
- 3 Regression estimates the average conditional relationship. OLS minimises the sum of squared residuals.
- 4 Simple regression shows raw association. Multiple regression shows partial associations (*ceteris paribus*).
- 5 Assumptions and diagnostics are not optional — they determine whether results can be trusted.

EXIT TICKET — 60 Seconds